

Data Communication

Tutorial 1 –

Chapter 1

- 1) Why data communication is important?
- 2) Draw and explain a generic data communication model. Explain in brief the basic signal processing operations at each block.
- 3) Explain the five essential components of a data communication system. How does each component contribute to successful communication? Explain with suitable diagram.
- 4) How is digital transmission superior to analog transmission system?
- 5) Explain digital communication system with its block diagram.
- 6) Explain the applications of data communication.
- 7) How data is transferred from sender to receiver in digital communication? Explain with necessary components. (*Hint: Show source encoder/ decoder, channel encoder/ decoder also*)

Chapter 2

- 8) Define Baud Rate. Explain RS 232C standards.
- 9) Explain data frame format of synchronous and asynchronous communication and differentiate them.
- 10) Differentiate between serial and parallel transmission.
- 11) Define channel capacity with necessary theorem. For a RF signal with bandwidth 6 MHz and it is transmitted with average power 2W where 1.2mW noise signal is present on it. If the signal is digitized and sent find the maximum data rate of the channel.
- 12) Explain different modes of serial transmission.
- 13) Explain the concept of bit rate, baud rate. Show their relationship.
- 14) If a communication channel has a bandwidth of 3 kHz and SNR of 31 dB, what would be the channel capacity using Shannon capacity theorem?
- 15) Explain transmission modes in data communication.
Hint: So when it comes to data transmission in computer networks, we define modes by the direction of data flow (simplex, half-duplex, full duplex), the number of simultaneous bits being sent (serial or parallel), and the synchronicity of devices (synchronous, asynchronous, isochronous). Explain all these.